

Question

Pain and motor severity
co-occur in PD

Which comes first?
Or neither?

Dopaminergic dose
changes with both

Method

RI-CLPM
separates between-person
from within-person

Mixed model + IPCW
level vs slope,
attrition weighted

MSM with stabilised IPTW
time-varying confounding
by prior exposure

Finding

Stable trait covariation
 $r = 0.152$
No temporal precedence

Level effect, not slope
+1.396 points/pain point
Slope $p = 0.189$

Attenuates by 37%
but persists
1.706 (0.717 to 2.670)